Principle of Virtual Work

Introduction

The principle of virtual work, although it is mentioned in almost every mechanics textbook, has been taught and used without properly explaining the physics and mathematics behind it.

Instead of deriving from the so-called "weak form" of equilibrium equation, which serves as a mathematical approximation technique of weakening the point-by-point restriction over a wider region (we shall discuss this technique in later chapter), the principle of virtual work is the exact variation of the energy principles (not unique). The result of two ideas can be identical in mathematical expression but can be very different when taking further steps of implementation.

These energy principles can be single field principles:

- Displacement Field – Potential Strain Energy – Principle of Virtual Displacement
- Stress Field – Complementary Strain Energy – Principle of Virtual Stress

These two principles are probably the most physical principles and construct the principle of virtual work in a narrow sense. We shall discuss these two principles here in this chapter and illustrate the famous Castigliano's theorem as a direct application of the two principles in 1-D element system in the later chapter.

They can also be multifield principles:

- Stress & Displacement Field – Hellinger Reissner Functional
- Stress, Strain, & Displacement Field – Hu Washizu Functional

These two principles are raised mainly for numerical purposes and are difficult to understand in an intuitive way. We shall discuss these two principles in the later chapter.

There can certainly be more principles by constructing among stress, strain, and displacement fields but are mainly mathematical curiosities.

Virtual Work under Virtual Displacement

Recall the derivation in the potential & complementary strain energy chapter, the potential strain energy and its variation:

$$V_\epsilon(\boldsymbol{\epsilon}) = \int v_\epsilon(\boldsymbol{\epsilon})\, d\Omega$$

$$\delta V_\epsilon(\boldsymbol{\epsilon}) = \int \boldsymbol{\sigma}(\boldsymbol{\epsilon}) : \boldsymbol{\delta\epsilon}\, d\Omega = \int \boldsymbol{\delta u} \cdot \boldsymbol{\sigma}(\boldsymbol{\epsilon}) \cdot \boldsymbol{n}\, d\Gamma_h + \int \boldsymbol{\delta u} \cdot \boldsymbol{q}\, d\Gamma_q + \int \boldsymbol{\delta u} \cdot \boldsymbol{f}\, d\Omega$$

We can see that the terms in right hand side are associated with external forces (stress field times virtual displacement). Further define the work done by external forces:

$$W_{u,ext}(\boldsymbol{u}) = \int \boldsymbol{u} \cdot \boldsymbol{\sigma}(\boldsymbol{\epsilon}) \cdot \boldsymbol{n}\, d\Gamma_h + \int \boldsymbol{u} \cdot \boldsymbol{q}\, d\Gamma_q + \int \boldsymbol{u} \cdot \boldsymbol{f}\, d\Omega$$

$$\delta W_{u,ext}(\boldsymbol{u}) = \int \boldsymbol{\delta u} \cdot \boldsymbol{\sigma}(\boldsymbol{\epsilon}) \cdot \boldsymbol{n}\, d\Gamma_h + \int \boldsymbol{\delta u} \cdot \boldsymbol{q}\, d\Gamma_q + \int \boldsymbol{\delta u} \cdot \boldsymbol{f}\, d\Omega$$

Since strain is a function of displacement:

$$\boldsymbol{\epsilon}(\boldsymbol{u}) = \frac{1}{2}(\boldsymbol{u} \otimes \boldsymbol{\nabla} + \boldsymbol{\nabla} \otimes \boldsymbol{u})$$

We can reconstruct the potential strain energy functional $V_\epsilon(\boldsymbol{\epsilon}) \Longrightarrow V_\epsilon(\boldsymbol{u})$ and define the new functional:

$$\mathcal{L}_\epsilon(\boldsymbol{u}) = V_\epsilon(\boldsymbol{u}) - W_{u,ext}(\boldsymbol{u})$$

$$\delta \mathcal{L}_{\boldsymbol{\epsilon}}(\boldsymbol{u}) = \int \boldsymbol{\sigma}(\boldsymbol{u}) : \boldsymbol{\delta\epsilon}\, d\Omega - \int \boldsymbol{\delta u} \cdot \boldsymbol{\sigma}(\boldsymbol{u}) \cdot \boldsymbol{n}\, d\Gamma_h - \int \boldsymbol{\delta u} \cdot \boldsymbol{q}\, d\Gamma_q - \int \boldsymbol{\delta u} \cdot \boldsymbol{f}\, d\Omega = 0$$

Similar to the operation in Euler-Lagrange Equation, set $\boldsymbol{\delta u} = \boldsymbol{0}$ on Γ_h where $\boldsymbol{u}$ is prescribed:

$$\delta \mathcal{L}_\epsilon(\boldsymbol{u}) = \int \boldsymbol{\sigma}(\boldsymbol{u}) : \boldsymbol{\delta\epsilon}\, d\Omega - \int \boldsymbol{\delta u} \cdot \boldsymbol{q}\, d\Gamma_q - \int \boldsymbol{\delta u} \cdot \boldsymbol{f}\, d\Omega = 0$$

This equation is called the principle of virtual work under virtual displacement (principle of virtual displacement). The new functional $\mathcal{L}_\epsilon(\boldsymbol{u}) = V_\epsilon(\boldsymbol{u}) - W_{ext}(\boldsymbol{u})$ is the Lagrangian of the system, and the equilibrium is rewritten as the extremum of the Lagrangian.

Virtual Work under Virtual Stress

Recall the derivation in the potential & complementary strain energy chapter, the complementary energy and its variation:

$$V_c(\boldsymbol{\sigma}) = \int v_c(\boldsymbol{\sigma})\, d\Omega$$

$$\delta V_c(\boldsymbol{\sigma}) = \int \boldsymbol{\epsilon}(\boldsymbol{\sigma}) : \boldsymbol{\delta\sigma}\, d\Omega = \int \boldsymbol{h} \cdot \boldsymbol{\delta\sigma} \cdot \boldsymbol{n}\, d\Gamma_h + \int \boldsymbol{u}(\boldsymbol{\sigma}) \cdot \boldsymbol{\delta\sigma} \cdot \boldsymbol{n}\, d\Gamma_q - \int \boldsymbol{u}(\boldsymbol{\sigma}) \cdot div(\boldsymbol{\delta\sigma})\, d\Omega$$

We can see that the terms in right hand side are associated with external displacements (displacement field times virtual stress). Further define the work done by external displacement:

$$W_{\sigma,ext}(\boldsymbol{\sigma}) = \int \boldsymbol{h} \cdot \boldsymbol{\sigma} \cdot \boldsymbol{n}\, d\Gamma_h + \int \boldsymbol{u} \cdot \boldsymbol{\sigma} \cdot \boldsymbol{n}\, d\Gamma_q - \int \boldsymbol{u}(\boldsymbol{\sigma}) \cdot [div(\boldsymbol{\sigma}) + \boldsymbol{f}]\, d\Omega$$

Note that:

$$\int \boldsymbol{u}(\boldsymbol{\sigma}) \cdot [div(\boldsymbol{\sigma}) + \boldsymbol{f}]\, d\Omega = 0$$

$$\delta W_{\sigma,ext}(\boldsymbol{\sigma}) = \int \boldsymbol{h} \cdot \boldsymbol{\delta\sigma} \cdot \boldsymbol{n}\, d\Gamma_h + \int \boldsymbol{u}(\boldsymbol{\sigma}) \cdot \boldsymbol{\delta\sigma} \cdot \boldsymbol{n}\, d\Gamma_q - \int \boldsymbol{u}(\boldsymbol{\sigma}) \cdot div(\boldsymbol{\delta\sigma})\, d\Omega$$

Define the new functional:

$$\mathcal{L}_c(\boldsymbol{\sigma}) = V_c(\boldsymbol{\sigma}) - W_{\sigma,ext}(\boldsymbol{\sigma})$$

$$\delta\mathcal{L}_c(\boldsymbol{\sigma}) = \int \boldsymbol{\epsilon}(\boldsymbol{\sigma}) : \boldsymbol{\delta\sigma}\, d\Omega - \int \boldsymbol{h} \cdot \boldsymbol{\delta\sigma} \cdot \boldsymbol{n}\, d\Gamma_h - \int \boldsymbol{u}(\boldsymbol{\sigma}) \cdot \boldsymbol{\delta\sigma} \cdot \boldsymbol{n}\, d\Gamma_q + \int \boldsymbol{u}(\boldsymbol{\sigma}) \cdot div(\boldsymbol{\delta\sigma})\, d\Omega$$

Similar to the operation in Euler-Lagrange Equation, set $\boldsymbol{\delta\sigma} = \boldsymbol{0}$ on Γ_q where $\boldsymbol{\sigma} \cdot \boldsymbol{n} = \boldsymbol{q}$ is prescribed and in Ω where $\boldsymbol{f}$ is prescribed:

$$\delta\mathcal{L}_c(\boldsymbol{\sigma}) = \int \boldsymbol{\epsilon}(\boldsymbol{\sigma}) : \boldsymbol{\delta\sigma}\, d\Omega - \int \boldsymbol{h} \cdot \boldsymbol{\delta\sigma} \cdot \boldsymbol{n}\, d\Gamma_h$$

This equation is called the principle of virtual work under virtual stress (principle of virtual stress). The new functional $\mathcal{L}_c(\boldsymbol{\sigma}) = V_c(\boldsymbol{\sigma}) - W_{\sigma,ext}(\boldsymbol{\sigma})$ is the Lagrangian of the system, and the equilibrium is rewritten as the extremum of the Lagrangian.

Global Rigid Body Equilibrium of a Deformable Body

The principle of virtual displacement can help us verify the deformable body shares the same equilibrium as rigid body when viewed globally.

Rigid Body Equilibrium:

$$\Sigma\boldsymbol{F} = \boldsymbol{0} \;\Leftrightarrow\; \begin{cases} \Sigma F_x = 0 \\ \Sigma F_y = 0 \\ \Sigma F_z = 0 \end{cases}$$

$$\Sigma\boldsymbol{M}_o = \Sigma\boldsymbol{r}_o \times \boldsymbol{F} = \boldsymbol{0} \;\Leftrightarrow\; \begin{cases} \Sigma M_{ox} = 0 \\ \Sigma M_{oy} = 0 \\ \Sigma M_{oz} = 0 \end{cases}$$

Principle of Virtual Displacement (full expression):

$$\delta\mathcal{L}_{\boldsymbol{\epsilon}}(\boldsymbol{u}) = \int \boldsymbol{\sigma}(\boldsymbol{u}) : \boldsymbol{\delta\epsilon}\, d\Omega - \int \boldsymbol{\delta u} \cdot \boldsymbol{\sigma}(\boldsymbol{u}) \cdot \boldsymbol{n}\, d\Gamma_h - \int \boldsymbol{\delta u} \cdot \boldsymbol{q}\, d\Gamma_q - \int \boldsymbol{\delta u} \cdot \boldsymbol{f}\, d\Omega = 0$$

Since $\boldsymbol{\delta u}$ can be any function that has first order continuity C^1, select:

$$\boldsymbol{\delta\epsilon}^* = \frac{1}{2}(\boldsymbol{u}^* \otimes \boldsymbol{\nabla} + \boldsymbol{\nabla} \otimes \boldsymbol{u}^*) = \boldsymbol{0}$$

$$\delta\mathcal{L}_{\boldsymbol{\epsilon}}(\boldsymbol{u}) = -\int \boldsymbol{\delta u}^* \cdot \boldsymbol{\sigma}(\boldsymbol{u}) \cdot \boldsymbol{n}\, d\Gamma_h - \int \boldsymbol{\delta u}^* \cdot \boldsymbol{q}\, d\Gamma_q - \int \boldsymbol{\delta u}^* \cdot \boldsymbol{f}\, d\Omega = 0$$

The two sets of vectors $\boldsymbol{\delta u}^*$ that satisfy the special requirement $\boldsymbol{\delta\epsilon}^* = \boldsymbol{0}$:

Set 1:

$$\boldsymbol{\delta u}^* \otimes \boldsymbol{\nabla} = -\boldsymbol{\nabla} \otimes \boldsymbol{\delta u}^* = \boldsymbol{0}$$

$$\boldsymbol{c} = c_i \boldsymbol{e}_i \quad c_i \in R \text{ is constant}$$

$$\boldsymbol{\delta u}^* = \boldsymbol{c}$$

$$\boldsymbol{c} \otimes \boldsymbol{\nabla} = \mathbf{c} \otimes \boldsymbol{\nabla} = \boldsymbol{0}$$

$$\boldsymbol{\delta\epsilon}^* = \frac{1}{2}(\boldsymbol{c} \otimes \boldsymbol{\nabla} + \boldsymbol{\nabla} \otimes \boldsymbol{c}) = \boldsymbol{0}$$

$$\delta\mathcal{L}_{\boldsymbol{\epsilon}}(\boldsymbol{u}^*) = \boldsymbol{c} \cdot \left[-\int \boldsymbol{\sigma}(\boldsymbol{u}) \cdot \boldsymbol{n}\, d\Gamma_h - \int \boldsymbol{q}\, d\Gamma_q - \int \boldsymbol{f}\, d\Omega\right] = \boldsymbol{0} \;\Leftrightarrow\; \Sigma \boldsymbol{F} = \boldsymbol{0}$$

Set 2:

$$\boldsymbol{u}^* \otimes \boldsymbol{\nabla} = -\boldsymbol{\nabla} \otimes \boldsymbol{u}^* \neq \boldsymbol{0}$$

$$\boldsymbol{\omega} = \omega_i \boldsymbol{e}_i \quad \omega_i \in R \text{ is constant}$$

$$\boldsymbol{X} = X_i \boldsymbol{e}_i \quad X_i \in R \text{ is the Cartesian coordinate based on any origin}$$

$$\boldsymbol{u}^*(\boldsymbol{X}) = \boldsymbol{\omega} \times \boldsymbol{X}$$

$$(\boldsymbol{\omega} \times \boldsymbol{X}) \otimes \boldsymbol{\nabla} = \omega_i \delta_{jl} \epsilon_{ijk} \boldsymbol{e}_k \otimes \boldsymbol{e}_l = -\omega_i \delta_{jl} \epsilon_{ijk} \boldsymbol{e}_l \otimes \boldsymbol{e}_k = -\boldsymbol{\nabla} \otimes (\boldsymbol{\omega} \times \boldsymbol{X})$$

$$\boldsymbol{\delta\epsilon}^* = \frac{1}{2}[(\boldsymbol{\omega} \times \boldsymbol{X}) \otimes \boldsymbol{\nabla} + \boldsymbol{\nabla} \otimes (\boldsymbol{\omega} \times \boldsymbol{X})] = \boldsymbol{0}$$

Note:

$$\boldsymbol{a} \times \boldsymbol{b} \cdot \boldsymbol{c} = \boldsymbol{a} \cdot \boldsymbol{b} \times \boldsymbol{c}$$

$$
\begin{aligned}
\delta\mathcal{L}_{\boldsymbol{\epsilon}}(\boldsymbol{u}^*) &= -\int (\boldsymbol{\omega}\times\boldsymbol{X})\cdot\boldsymbol{\sigma}(\boldsymbol{u})\cdot\boldsymbol{n}\,d\Gamma_h - \int (\boldsymbol{\omega}\times\boldsymbol{X})\cdot\boldsymbol{q}\,d\Gamma_q - \int (\boldsymbol{\omega}\times\boldsymbol{X})\cdot\boldsymbol{f}\,d\Omega \\
&= \boldsymbol{\omega}\cdot\left[-\int \boldsymbol{X}\times\boldsymbol{\sigma}(\boldsymbol{u})\cdot\boldsymbol{n}\,d\Gamma_h - \int \boldsymbol{X}\times\boldsymbol{q}\,d\Gamma_q - \int \boldsymbol{X}\times\boldsymbol{f}\,d\Omega\right]\mathbf{0} \;\Leftrightarrow\; \Sigma\boldsymbol{M}_o = \Sigma\boldsymbol{r}_o\times\boldsymbol{F} \\
&= \mathbf{0}
\end{aligned}
$$